David Lewis

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EDUCATION

Full Time Student, The University of Texas at Dallas Richardson, TX
B.S., Computer Science Anticipated Dec 2026
GPA: 3.82
Relevant Coursework: Computer Science, Coding, **Data Structures**, Algorithms, **Mathematics**, Professional Communication
Honors: **National Merit Finalist** Scholarship Recipient, UTD Computing Scholars Honors Program **(CS^2)**, UTD CV Honors

SKILLS

Languages: **Java**, **Python**, C, C++, MATLAB, **TypeScript**, JavaScript
Frameworks & Tools: **Git**, Docker, **Linux**, TailwindCSS, **React**, **OOP**, Multithreading, **APIs**, MongoDB
Robotics & AI: **Robot Operating System 2 (ROS2)**, SLAM, YOLO,
Additional: Communication, **Leadership**, Presentations

COLLABORATIVE PROJECTS

UT Dallas ACM
DevDiary Projects Developer Sep 2025 – Present
- Building a full-stack documentation and debugging assistant with team of four using **MERN** stack
- Architecting an **auto-generated tagging** system to automatically summarize and index logs
- Implementing an **AI-assisted workflow** integrating OpenAI APIs to explain errors and provide debugging suggestions
- Designed and optimized **MongoDB schemas** for customizable logs and projects to ensure low-latency queries.

UT Dallas RoboSub
RoboBoat Programming Lead Aug 2024 – Present
- Leading a programming team in developing an **Autonomous** Surface Vehicle (**ASV**) for international competition
- **Mapping** course and **localizing** autonomous boat with **SLAM** and **Loop Closure Detection** Algorithms.
- Implementing autonomy with **Python** and **ROS2** by planning an **RTP** based path, achieving a 70% success rate
- Mentoring team members in **Git**, **Docker**, modular software design, and cross-discipline collaboration

UT Dallas BioTech Club
SmartPlate Team Lead Feb 2025 – May 2025
- Directed development of an **AI** driven smart plate to track nutrient intake in collaboration with a biotech startup
- Developed a **Computer Vision pipeline** that identifies food items using a trained YOLOv8 model
- Integrated sensor readings from load cells with vision model predictions to compute nutrient values in real time

PERSONAL PROJECTS

Konbini Locator Jul 2025 – Aug 2025
- Built an analysis tool in **Python** integrating **Google Maps API** to find the tightest cluster of Konbinis in Tokyo
- Optimized performance with **multithreading** to **parallelize** 100+ **API** requests reducing data collection time by **70%**
- Designed a **R*-Tree** based **algorithm** to compute the minimum-area triangle made from stores in different chains

Simple Neural Network Jun 2024 – Jun 2024
- Independently studied the fundamentals of **neural networks**, backpropagation, and gradient descent
- Implemented a **Java**-based neural network to recognize handwritten digits using the MNIST dataset
- Gained hands-on experience in **machine learning**, including model training, evaluation, and tuning

Chess AI Jan 2023 – Mar 2023
- Designed and developed a fully functional graphical chess engine from the ground up in **Java**
- Engineered an **AI** opponent using the minimax algorithm with heuristic board evaluation and recursive decision making
- Achieved an ELO rating of 700 against real humans on chess.com

WORK EXPERIENCE

Tutor Arlington, TX
Kumon of North Arlington Sep 2021– Jun 2024
- **Tutored** 100+ students in **math** and reading with personalized feedback and tracked progress
- **Communicated** student performance trends to parents and center instructors, contributing to individual learning plans.